CSE 3031 Computer Architecture

Homework 4

Spring 2026

Deadline: Midnight of Thursday, Apr. 23

Problem 1 (20 pts, 5/15): Consider a pipelined processor with a 5-stage pipeline that is perfectly balanced. For simplicity, assume that all instructions take 5 cycles. The dynamic instruction count by type, as a percentage of the total, is as follows:
- 10% stores - 25% loads - 15% branches - 50% ALU
(a) What is the ideal speedup due to pipelining for this processor compared with unpipelined single-cycle processor? Explain.
(b) Stalls due to data hazards occur only under two scenarios. A stall of one cycle occurs when a load instruction is followed by an ALU instruction that uses the result of the load. This scenario exists for 40% of the load instructions. A stall of two cycles occurs when a branch instruction is preceded by an ALU operation whose result is used as the branch condition. This scenario exists for 50% of the branch instructions. What is the decrease in the ideal speedup of pipelining due to the two data hazards?

Problem 2 (20 pts, 5/5/5/5) 2.The instruction execution of a computer can be divided into five stages as follows: IF, ID, EX, MEM and WB. The measured times for the stages are: 1ns, 1.5ns, 1ns, 2ns, and 1.5ns, respectively.
(a) What is the clock cycle time if the instruction execution is implemented as a single-cycle implementation?
(b) What is the clock cycle time if the instruction execution is implemented as a multiple-cycle implementation?
(c) What is the clock cycle time if the instruction execution is implemented as a pipelined implementation?
(d) What is the speedup of the pipelined implementation over the single-cycle implementation?

Problem 3 (20 pts): For a deeper pipeline, such as that in MIPS R4000, it takes three pipeline stages before the branch target address is known and an additional cycle before the branch condition is evaluated (taken or not taken) (i.e. the branch target address is calculated at stage 3 and branch result is evaluated at stage 4). Assume no stalls on the registers in the conditional comparison.
Fill in the table for the number of cycles of penalty for the three static branch approaches. Note: if what you have fetched is correct, you do not need to repeat the instruction fetch.

Strategy	Unconditional Branch	Conditional Branch - Taken	Conditional Branch - Untaken
Stall the pipeline until the branch is resolved			
Predict taken			
Predict untaken			

Problem 4 (20 pts): Identify all data dependencies (RAW, WAR, WAW) in the code below. Assume a simple MIPS 5-stage pipeline is used (register write in the first half of a cycle and register read in the second half). Find the dependency that causes a data hazard and state whether it requires a stall to be handled, or can be handled via forwarding. If there is no data hazard, put "No" in the last column. Add extra rows to the table if needed.

```
1:      LD R1, 16(R2)          ; Reg[R1] <- Mem[Reg[R2]+16]
2:      DADD R7, R1, R8        ; Reg[R7] <- Reg[R1] + Reg[R8]
3:      DSUB R8, R1, R6        ; Reg[R8] <- Reg[R1] - Reg[R6]
4:      OR R1, R5, R3          ; Reg[R1] <- Reg[R5] OR Reg[R3]
5:      SD R7, 64(R2)          ; Reg[R7] -> Mem[Reg[R2]+64]
```

Instruction 1	Instruction 2	Register	Dependency	Stall/Forward

Problem 5 (20 pts): Consider a loop that is entered two times in sequence in a program. Each time it is entered, the loop performs 4 iterations. Each iteration executes two branches in sequence with the following outcomes:

Iteration	1	2	3	4	1	2	3	4
Branch 1	T	N	T	N	T	N	T	N
Branch 2	T	T	T	N	T	T	T	N

When Branch 2 is not taken at iteration 4, the program leaves the loop. Record the predictions of the three branches using the following two branch predictors: (1) a 2-bit dynamic up/down saturation counter predictor (initial state 00), (2) a (1,1) correlated predictor (initial state N/N). Assume that the outcome of the branch instruction executed before entering the loop is N.

(a) 2-bit saturation counter predictor

Branch 1

Iteration	1	2	3	4	1	2	3	4
Branch Outcome	T	N	T	N	T	N	T	N
Predictor State	00							
Prediction								

Branch 2

Iteration	1	2	3	4	1	2	3	4
Branch Outcome	T	T	T	N	T	T	T	N
Predictor State	00							
Prediction								

(b) (1, 1) correlated predictor

Branch 1

Iteration	1	2	3	4	1	2	3	4
Branch Outcome	T	N	T	N	T	N	T	N
Predictor State	N/N							
Prediction								

Branch 2

Iteration	1	2	3	4	1	2	3	4
Branch Outcome	T	T	T	N	T	T	T	N
Predictor State	N/N							
Prediction								

How to submit: Write your answer in a docx or pdf file. Upload the file through the submission link on Canvas.